

Introduction to Networking

We need to know the concept of 'NETWORKING' as it helps us to understand how data flows from source to destination with integrity.

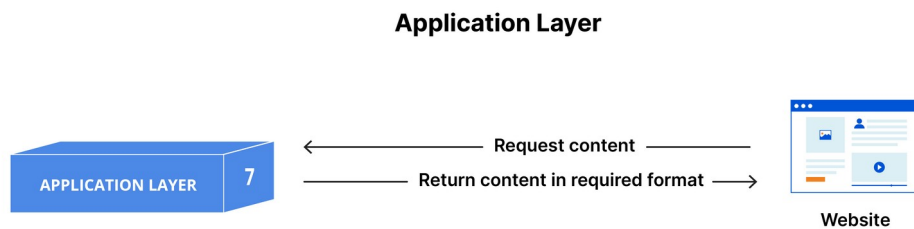
➤ OSI Model

The OSI Model is an approach to teach how data traverses. It is a theoretical model that provides easy understanding of data transmission process. Developed by ISO in 1978 and officially published in 1980. ISO is an International Standard Organisation that explained 7 layered OSI (Open System Interconnection) model for devices that are open to communication. It is only a theoretical model and not used practically.



OSI Model was created with layers (7 layers) so that each task can be assigned to a separate layer and tasks should not be assigned to the same layer again eliminating ambiguity and assuring security and integrity. Therefore each layer in the OSI Model has its separate task associated. We will gain insight of each layer now.

Application Layer (7th layer)



It is the last layer of the OSI Model. It is an interface between the user and application. It is the only layer that is directly in contact to the user. What you see on your screen I.e applications on the screen is the application layer of the networking model. Each layer uses multiple protocols. The protocol used by this layer are HTTP, HTTPS, DNS, FTP etc.

Presentation Layer



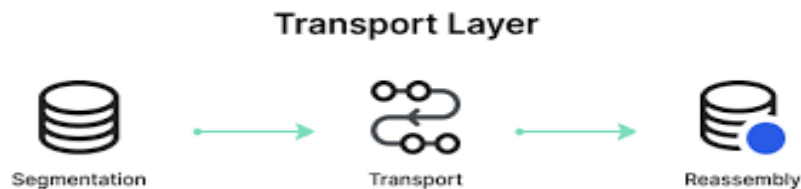
Presentation layer is the 6th layer of OSI model. It is used to encrypt-decrypt, compress and secure the data from Application layer. It uses protocols like SSL/DTLS, XOR, NDR, MIME etc.

Session Layer



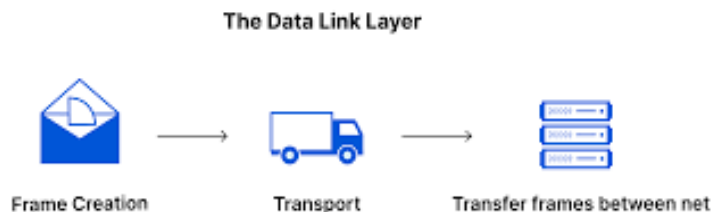
Session layer is 5th layer in OSI Model. It is responsible for establishing, managing and terminating the communication sessions in the network. NetBIOS is a protocol used by this layer.

Transport Layer



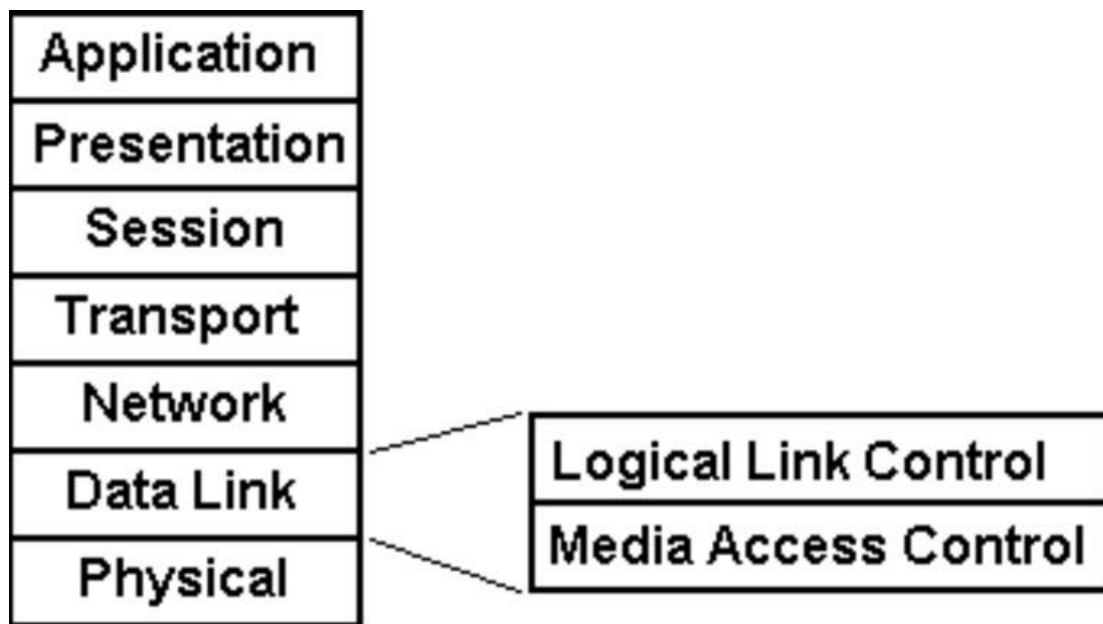
Transport layer is the 4th layer in OSI model and as its name suggest it works with transmission of the data to its destination. It connects the network layer to all the above layers in the network. It uses protocols like TCP (segments) and UDP (datagram) to transmit the data.

Network Layer



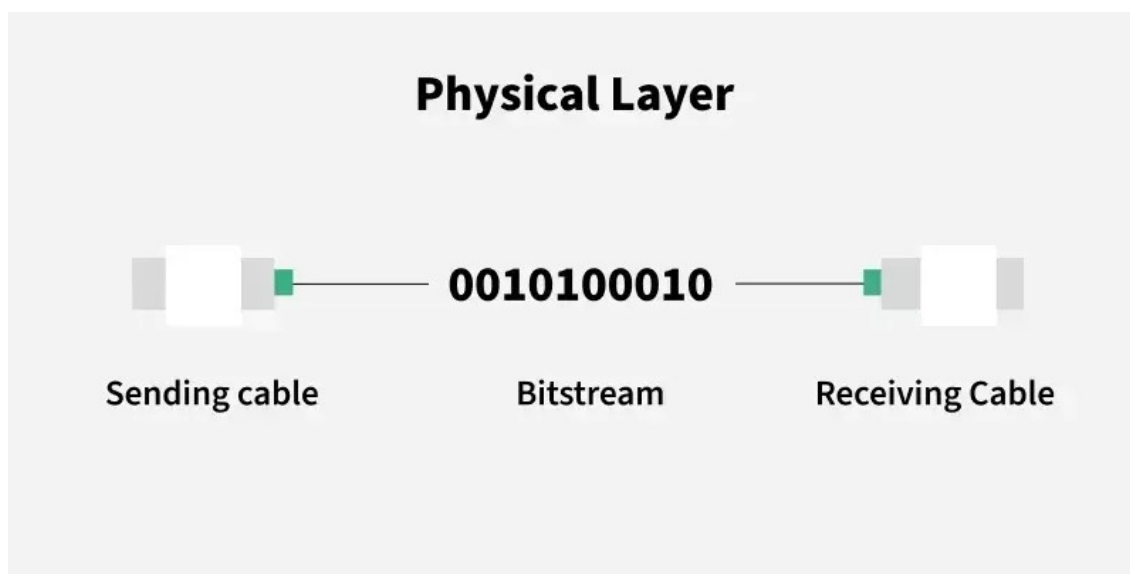
Network layer is the 3rd and the most essential layer (one of the most essential layer) in the model. It divides the data segments from transport layer into data frames and also reassembles them when necessary. It assigns logical addressing i.e. IP address to the devices in the network. It specifies the routing paths from tran for transmission of the data. Routers are the devices that operate at this layer.

Data Link Layer



Data link layer is the 2nd layer of the OSI Model. It is divided into two parts i.e. Data Link Control and MAC (Media Access Control) each having their separate tasks. The DLC (Data Link Control) provides a connection between the Network layer and MAC layer. Also controls the data flow, Error Detection and Framing. The MAC layer interact directly with the hardware i.e. Ethernet cables, fibres, wires etc. Also provides the Physical (MAC) address to the device through NIC cards and frames the data for transmission. Switches are the devices that operate at the Data link layer. The Data Link Control (DLC) controls the medium of transmission's access and does the multiplexing and demultiplexing of the communication channel.

Physical Layer



It is the lowermost i.e 1st layer of the OSI Model. It is the hardware that helps in transmission of the data. For example Ethernet cables, wires, fibers etc. It manages the flow of data and transmits it as a bit stream over the physical medium. It makes sure error free transmission of data by overseeing the bit is received as it is transmitted. i.e. If 1 is received as 1 by the receiver. It supports point to point and Point to multipoint connection between devices.

Each layer adds and removes its headers as data passes through each of them. It then encapsulates them into their data packets and passed them to the next layer.

